Quantifying the Decadal, Annual, and Seasonal Trends of Sedimentation Rate on the Shoreface of Emerald Isle, North Carolina

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ENEC 395

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December 11th, 2024

Abstract

Emerald Isle, North Carolina, is a town on Bogue Banks - a barrier island situated just south of Cape Lookout. This study specifically analyzed the shoreface of the westernmost end of the town, commonly referred to as “the point”, which is intersected by Bogue Inlet. Prior research within this area has concluded that there is a wide array of geologic factors that influence the bathymetry and topography seen today. Some factors include: the hard substrate underlying the current island position *(Riggs et al., 2001)*, as well as features such as ancient dune fields *(Riggs et al., 2001)* and paleochannels *(Timmons et al., 2010)*. Additionally, the area has been monitored and surveyed by the Carteret beach commission over the last 3 decades. Their findings show variance in annual change along specific regions of shoreline and focus in on management implications for renourishment projects (*Monitoring | Carteret County, NC - Official Website*, 2020). Using data collected from the Carteret County Beach Commission (CCSPO), a volumetric analysis was conducted along multiple areas of interest to quantify the sedimentation trends of the area between the years 2013 and 2024. Taking a multifaceted temporal approach to data analysis, the inter-seasonal, inter-annual, and decadal trends of the beach strand were analyzed and compared. Results showed that there has been a significant net accretion of sediment on the point’s beach strand over the last decade and provided insight to the variance during all measured time periods in reaction to constant natural pressures such as wave energy and nearshore currents. Possible mechanisms for the general trend of accretion over the last decade were proposed and tested, however the results are still inconclusive to the source of net accretion over the last ten years. These analyses aimed to provide better insight on long term sedimentation trends on the Point of Emerald Isle. By doing so, the study found a multitude of

seasonal, annual, and decadal patterns, and outlined future research ideas to attempt to better understand the abnormality of the area.

Introduction

Bogue Banks is a barrier Island situated in Southeastern North Carolina that is geologically referred to as “complex” due to its high elevation and wide topographical nature. (Riggs et al., 2001). The establishment of its topographical profile can be attributed to the dominant substrate which consists of very old, hard rocks that are a part of the “Carolina Platform” – a geologic feature stemming from the areas Cretaceous-Miocene geologic framework. *(Riggs et al, 2001)*. Due to the island’s elevation and width, oceanic overwash is limited moving from the Atlantic Ocean to Bogue Sound; Meaning estuarine-ocean sediment interaction is defined by inlet dynamics. *(Riggs et al. 2001)* The western end of Bogue Banks is intersected by Bogue Inlet, which is regularly traveled for recreational and commercial purposes.

Emerald Isle, North Carolina, is a town that encompasses the westernmost 13 miles of Bogue Banks. Originally purchased in 1954 for $350,000, Emerald Isle has slowly been developed over the last 7 decades into the fourth largest oceanfront municipality in North Carolina. (*Emerald Isle, NC | Official Website*, n.d.) As development has overcome the island and almost every oceanfront lot has some form of residential development, the fight against shoreline erosion is now in full swing.

The Carteret County Shore Protection Office (CCSPO) oversees local monitoring and preservation plans for the entirety of Bogue Banks. They currently have projects that include reoccurring, "touch up", and emergency dredging and renourishment. Reoccurring renourishment

and dredging take place on 3-to-4-year cycles. The cycles include timelines for monitoring and surveying. Touch up activities take place on yearly scales and are meant to reenforce precarious stretches of beach strand, inlets, and the intercoastal waterway. Emergency activities take place after major storm events such as Hurricane Florence (2018) and Hurricane Irene (2012). The current monitoring project features bi-yearly analysis of the beach strand. Utilizing metrics such as volume change and shoreline profiles, areas of concern or "hot spots" have been identified along the island. These hotspots are measured against a trigger volume - the threshold under which properties along the beach would no longer be protected from a 25-year storm.

Situated in a geographical commonplace for tropical weather landfall and nor'easter development, Emerald Isle is vastly impacted by seasonal swings in predominant wind flow, longshore current direction, and mean wave direction. All these variables play a major role in the general sediment dynamics along the shoreline and inlet. Over the last decade, Bogue Banks has been significantly impacted by a few major tropical weather events including Hurricane Arthur (2014), Hurricane Matthew (2016) *Figure 1*, Hurricane Florence (2018) *Figure 2*, Hurricane Dorian (2019) *Figure 3*, and Tropical Storm Isais (2020). These events have been a major driver in shoreline change, especially within hotspots along the island. The CCSPO found that Hurricane Florence single handedly caused the loss of over 1.6 million cubic feet of sand on Emerald Isles westernmost 3 miles of territory *(Moffatt and Nichol, 2019)*. Although Major storms have compounding effects on long term erosion rates that cumulatively cause net erosion across the island, certain areas over the last ten years have still accreted sediment due to a multitude of natural and anthropogenic factors. One of these areas is the western most end of Emerald Isle, commonly referred to as "The Point". *(Figure 4)* This study focusses specifically

on this area, including the intersecting Bogue Inlet, to both quantify seasonal trends in beach strand volumetric change and investigate the mechanisms that may drive these patterns.

Figure 4: "The Point" of Emerald Isle along with neighboring Bogue Inlet and Bear Island. The outlined area represents the span of the data analyzed in this study.

Methods

Digital Elevation Models – Visualizing Change

ArcGIS pro was used in all mapping and data extraction methods. A set of twenty Digital Elevation Models (DEM's), consisting of data from 2014 to 2023, were collected from the Carteret County Shore Protection Office. The DEM's encompass 10 transects *(figure 5)* that are a part of the CCSPO monitoring program, meaning they are mapped consistently with similar methods and approach. DEM data collection surveys occur bi-annually, once in the spring and once in the fall. Consequently, analysis was set for annual volumetric change (Fall to Fall), winter volumetric change (fall to spring), and summer volumetric change (spring to fall). After all, 20 DEM's were compiled and symbolized, the "raster calculator" tool was used to generate raster images that depict the difference in elevation between two DEM's. (*Figure 6)*

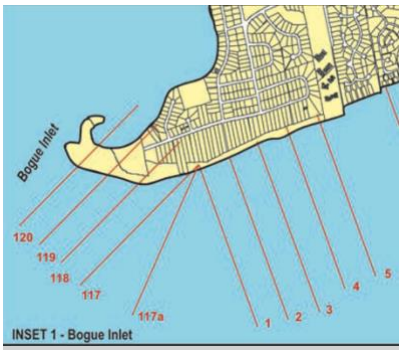

Figure 5: Transects showing the spatial scope of data analyzed in this study.

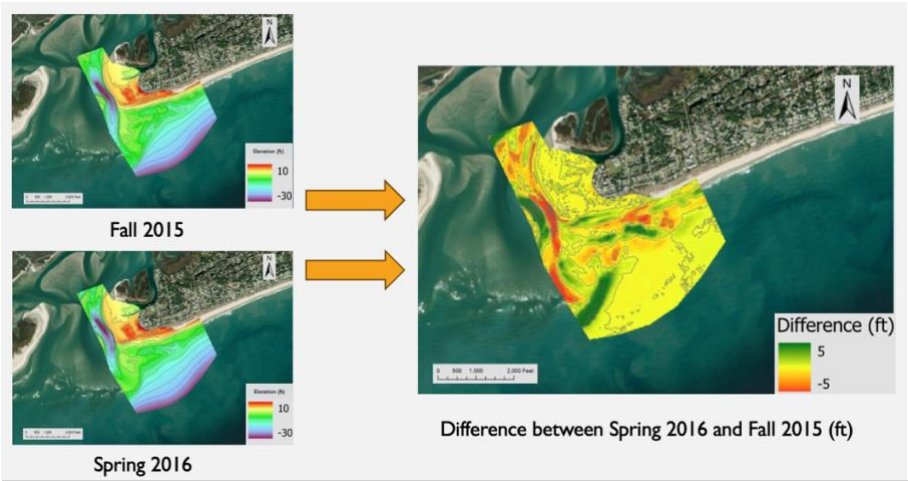

Figure 6: Raster Difference generation - Spring 2016 elevation is subtracted from Fall 2015 to create a model depicting the elevation difference between the two time periods.

This process was repeated for every annual and seasonal analysis window. Using the symbolized raster differences, elevation loss hotspots became visually identifiable. Visual analysis of the raster differences along with the base DEM's led to the creation of 3 initial Areas of Interest. The Beach Strand Zone – consisting of the entire shoreface - was designed to capture the general trends of the entire beach at the Point. Zone 1 was chosen for its observed tendency to accrete whereas Zone 2 was designed to encompass an area of general erosion. Both zones 1 and 2 were situated within the Beach Strand Zone so that accurate comparative analyses between all zones could take place. *(Figure 7)* Due to complications in data synthesis and analysis, Zone 2 will not be included in the current issue of this paper.

After analysis of the zones designated above, a fourth zone was generated to attempt to quantify a hypothetical mechanism for the seasonal trends in volumetric change seen on the beach strand. The Shoal Zone was designed to capture the volumetric trends of the neighboring shoals on the easterly side of Bogue Inlet. *(Figure 7)*

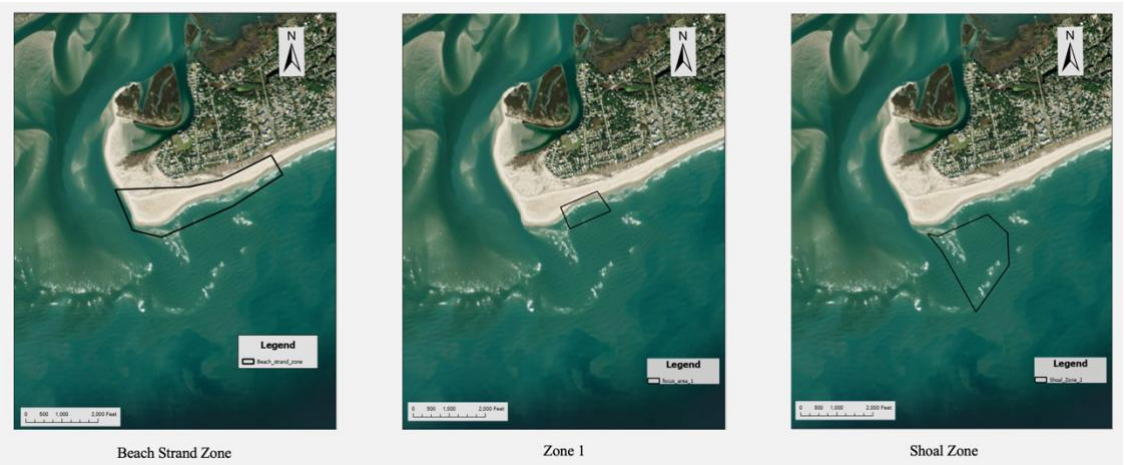

Figure 7: Images of the Beach Strand Zone, Zone 1, and the Shoal Zone.

Data Generation and Extraction

To generate data that is exportable into data analysis software, the "extract by mask" geoprocessing tool was used to extract the DEM's from within the AoI's. Following extraction, the "CutFill" Geoprocessing tool was used on the extracted DEM's *(figure 8).* CutFill organizes areas of net volumetric loss and gain in a way that allows for the transformation of data into a table that is extractable and formatted for external analysis.

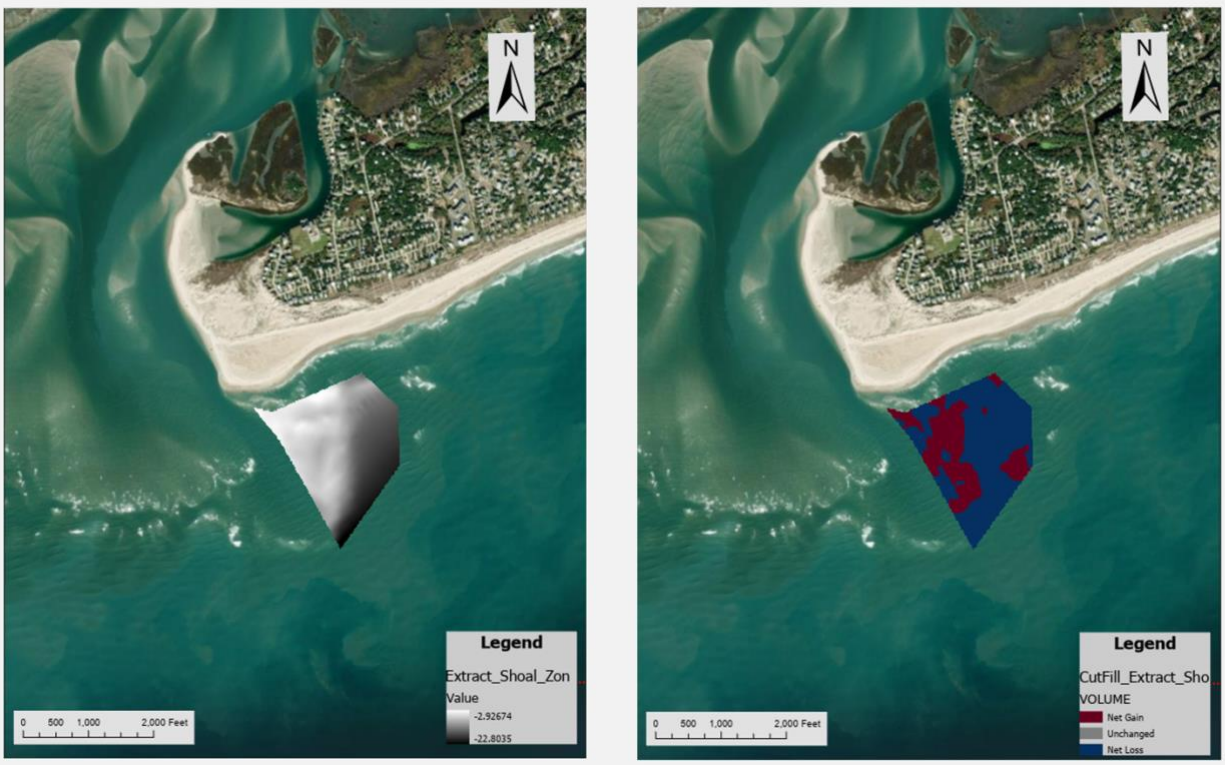

Figure 8: The "Extract by mask" and "CutFill" tools respectively. The DEM for a given survey is extracted through the area of interest. Following extraction, CutFill is used to give the difference between two extracted DEM's.

Data Analysis

All data extracted from ArcGIS was compiled into an Excel worksheet. Area and volume data were inputted from the Beach Strand Zone, Zone 1, Zone 2, and Shoal Zone for all years.

These values were then summed to get net volumetric change for all analysis periods (Decadal, annual, and seasonal). After summing data, the totals were cross compared, analyzed, and visualized for trends during any of the given time periods.

Results

Beach Strand Results: Decadal Overview

Over the last ten years, the Beach Strand Zone has accreted 9,000,000 (+/- 600,000) million cubic feet of sediment. The surface area of the zone from which the data was extracted is 4.21 million square feet, meaning the beach strand has gained an average of 2.34 feet of elevation. While there is a general trend of accretion throughout the data, the point does not accrete evenly seasonally. Roughly 95% of all accretion can be attributed to the summer months whereas only ~5% of the net accretion occurred during winter months. *(Figure 9)*

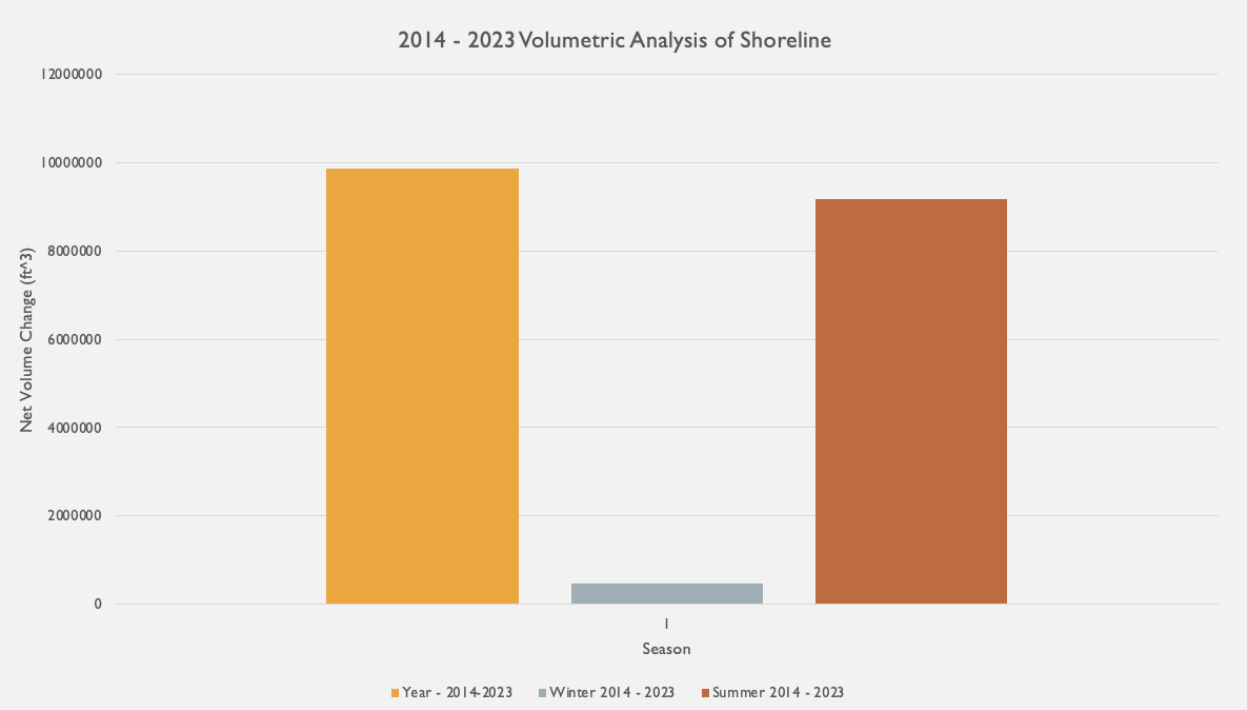

Figure 9: Beach Strand volumetric change from 2014 – 2023; Organized into net change, winter change, and summer change.

Beach Strand Results: Annual Breakdown

Decadal analysis provides insight into long term trends in beach strand dynamics. However, it does not provide insight into the variance that occurs annually during that period. Accretion did not occur evenly through time, rather there were periods of yearly net accretion and yearly net erosion. Over the last decade, the beach strand on the point net eroded for 4 years and net accreted 6 years. The largest total accretion occurred between the Fall of 2020 survey and the Fall of 2021 survey, when the area gained 4,750,000 (+/- 600,000) cubic feet of sediment. The largest erosive year occurred in 2015 when the beach strand lost a total of 2,310,000 (+/- 600,000) cubic feet. *(Figure 10)*

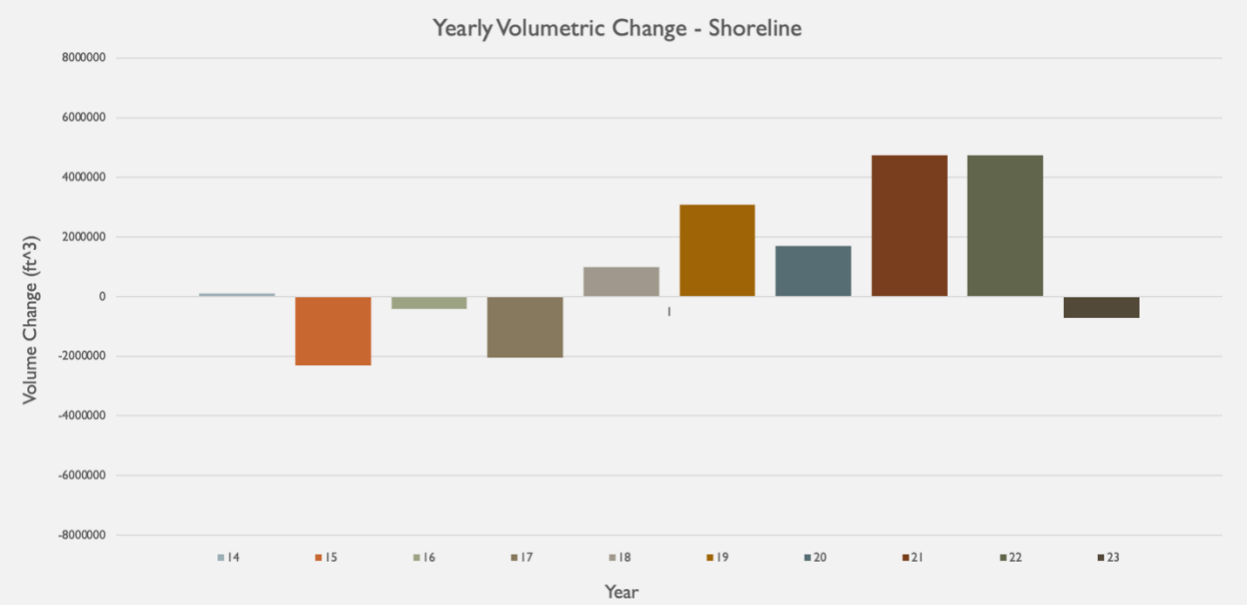

Figure 10: Beach Strand Zone annual volumetric change from 2014 - 2023

Beach Strand Results: Seasonal Analysis

Seasonal analysis of the beach strand provides more insight into how the area navigates erosional and accretional phases throughout the course of a year. Allowing for further analysis of the intra-annual trends that lead to long term sediment flux patterns. As seen in the annual volumetric change results, there is a large amount of year-to-year variance. Seasonal results suggest that variance increases further when the temporal scope is magnified to a shorter period. Within this study "winter months" are defined as the time between the fall and spring beach surveys and "summer months" are defined as the time between the spring and fall surveys. The change during these periods is calculated as the difference in volume between the two.

During the winter months, the beach strand cumulatively accreted 467,000 (+/- 600,000) cubic feet of sediment, accounting for 5.2% of the total accretion from 2014-2023. This relatively minor long-term rate of accretion is determined by an area that exhibits fluctuation in volumetric change over time. The Beach Strand Zone net accreted 6 winters, and net eroded 3 winters, with one winter representing a roughly net zero change. The largest change occurred in the winter of 2023 when the area lost a total of 7,130,000 (+/- 600,000) cubic feet, or an average elevation loss of -1.66 feet per square foot. *(Figure 11)*

During the summer months, the beach strand cumulatively accreted 8,600,000 (+/- 600,000) million cubic feet of sand, accounting for 95% of the total accretion from 2014-2023. Similar to the winter months, the beach strand eroded 3 years, accreted 6 years, and showed a roughly net zero change one year. While there was a congruency in number of years eroding and accreting, these years did not align. For example, the summer months demonstrated the highest rates of accretion in 2023 while the prior winter season showed the highest rates of erosion. During the summer months between the 2023 spring and fall surveys, the beach strand zone accreted a total

of 6,350,000 (+/- 600,000) cubic feet, gaining an average of 2.17 feet of elevation per square foot. *(Figure 11)*

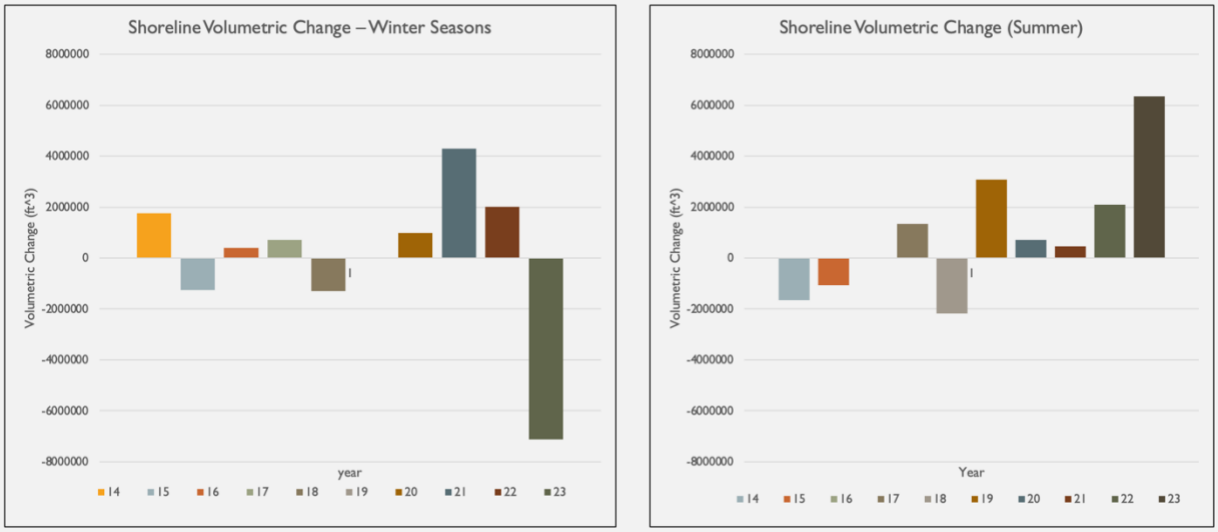

Figure 14: Area 1 Winter season and summer season volumetric change respectively from 2014 to 2023

Zone 1 Results: Decadal

Within the temporal scope of this study, Area 1 in total accreted 805,000 (+/- 300,000) cubic feet of sediment. The surface area of the polygon that the volumetric data was extracted from is 764,000 square feet, meaning that Area 1 has gained a total of 1.05 feet in elevation per square foot from 2014 to 2023. While there was a net accretion over the last decade, the change was not distributed evenly throughout time. All sediment gain within Area 1 can be attributed to the winter months, accreting a total of 1,920,000 (+/- 300,000) cubic feet. Contradictorily, the

summer months net eroded 1,100,000 (+/- 300,000) cubic feet, counteracting much of the deposition supplied in the winter. *(Figure 12)*

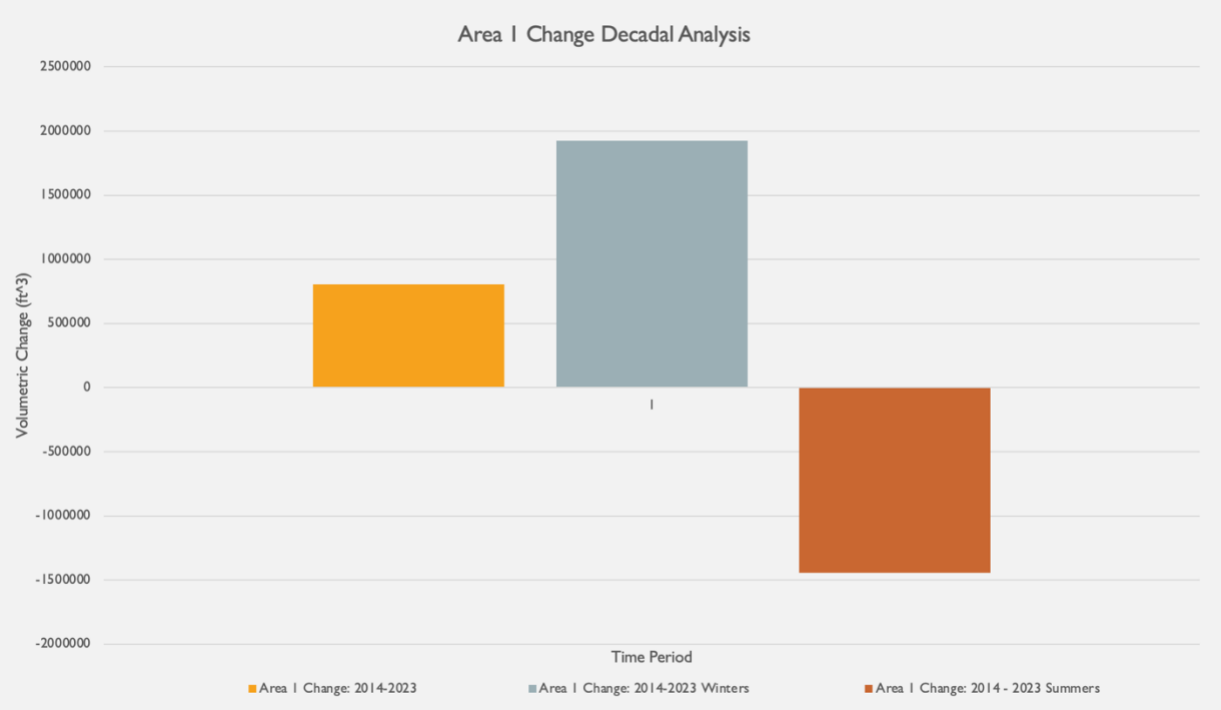

Figure 12: Area 1 volumetric change from 2014 – 2023; Organized into net change, winter change, and summer change respectively.

Zone 1 Results: Annual

Over the last decade, Zone 1 net eroded for 3 years, and net accreted 7 years. The largest total accretion occurred between the Fall of 2019 survey and the Fall of 2020 survey, when the zone gained 1,590,000 (+/- 300,000) cubic feet of sediment. Raising the elevation of the area on average by 2.09 feet per square foot. The largest erosive year occurred between the fall of 2022 and the fall of 2023 when the beach strand lost a total of ~2,150,000 (+/- 300,000) cubic feet, leading to an average change of -2.85 feet of elevation per square foot. *(Figure 13)*

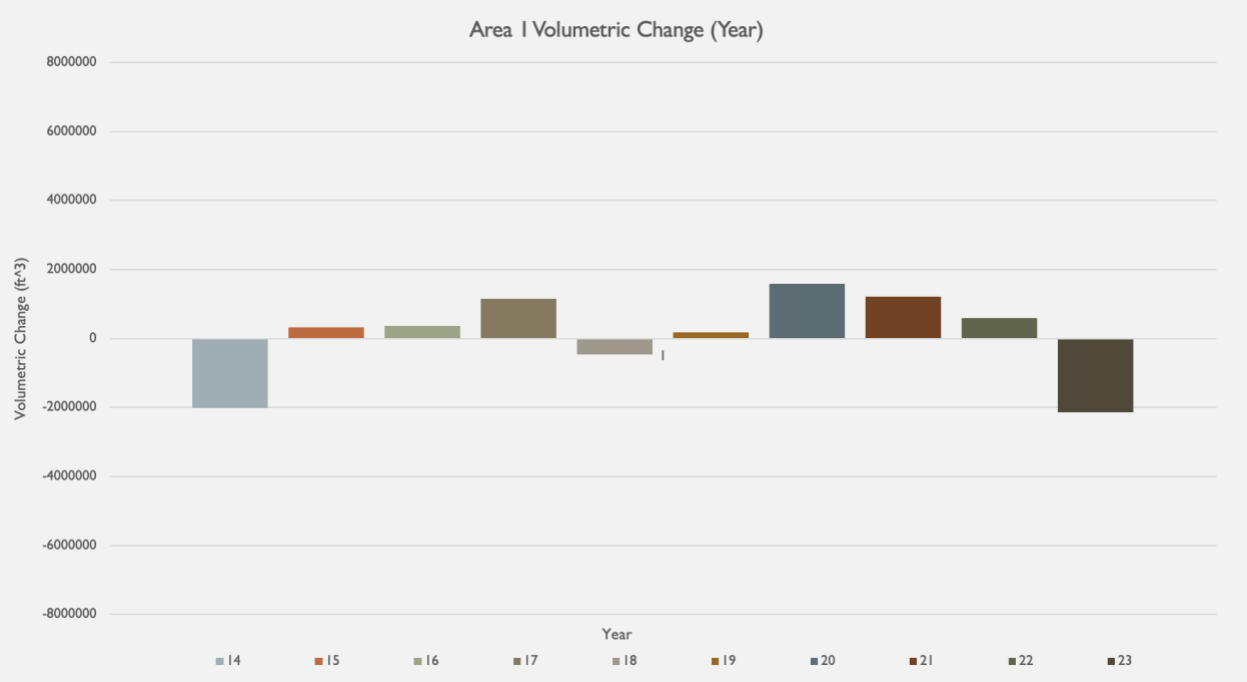

Figure 13: Zone 1 annual volumetric change from 2014 - 2023

Zone 1 results: Seasonal

During the winter months, Area 1 cumulatively accreted 1,920,000 (+/- 300,000) cubic feet of sediment, accounting for all the total accretion from 2014-2023. This relatively major long-term rate of accretion is determined by a period that exhibits extreme variance over time. Area 1 net accreted 6 winters, and net eroded 4 winters. Almost all the positive change occurred between the fall of 2019 and the fall of 2022. The largest change occurred in the winter of 2023 when the area lost a total of 2,100,000 (+/- 300,000) cubic feet, or an average elevation loss of -2.75 feet per square foot. *(Figure 14)*

During the summer months, Area 1 cumulatively eroded 1,440,000 (+/- 300,000) cubic feet of sediment, counteracting almost 75% of the accretion from the winter months. The year of

the greatest net change was from the spring of 2014 to the fall of 2014 when Area 1 eroded 1,900,000 (+/-300,000) cubic feet. *(Figure 14)*

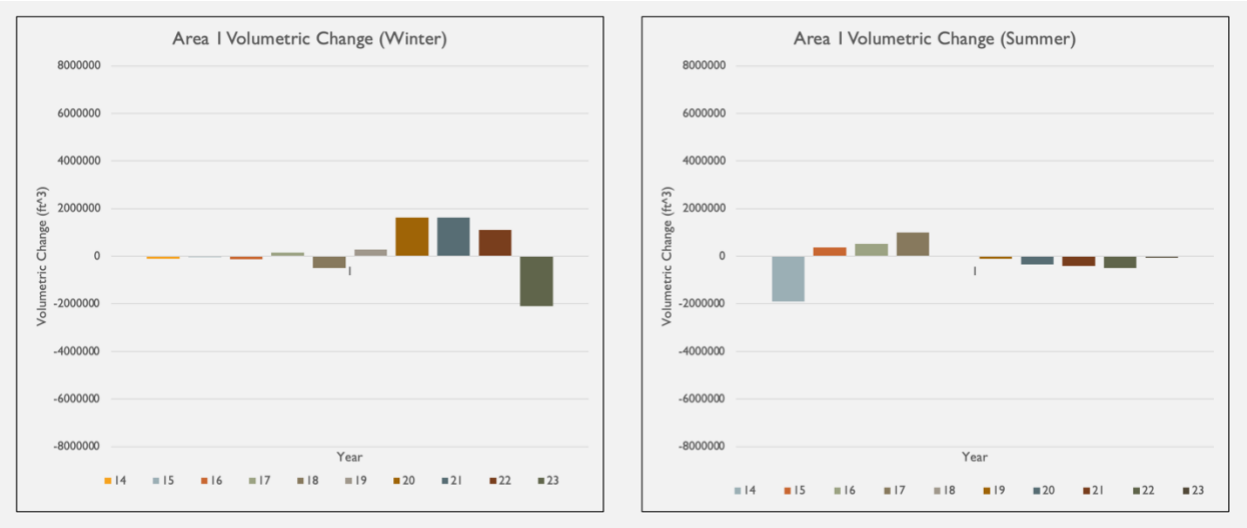

Figure 14: Area 1 Winter season and summer season volumetric change respectively from 2014 to 2023

Zone 2 Results

Area 2 data showed a significant amount of error upon further analysis and is currently excluded. Sources of this error are currently being investigated and hopefully resolved so that the results and discussion of those results can be written in the near future.

Shoal Zone Results: Decadal

The Shoal Zone was outlined and analyzed due to its possible role in the large-scale accretional patterns that were demonstrated in the Beach Strand Zone Results. Over the last 10 years, the nearshore area accreted more than the beach strand, gaining 12,000,000 (+/- 1,000,000) cubic feet of sediment, with the winter months accounting for 4,000,000 (+/- 1,000,000) cubic feet of gain and the summer months accounting for 6,500,000 (+/- 1,000,000) cubic feet of gain. Significantly more *(Figure 15)*

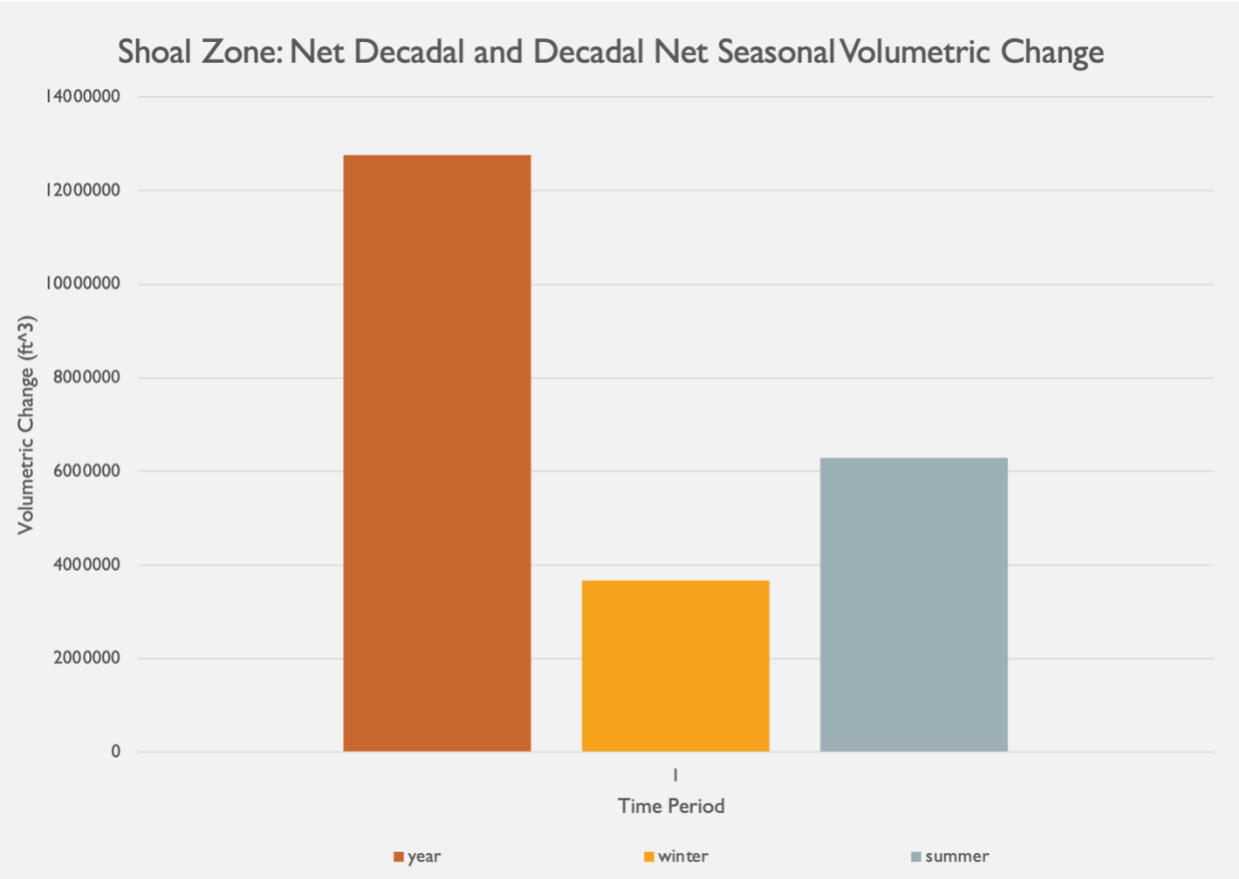

Figure 15: Area 1 volumetric change from 2014 – 2023; Organized into net change, winter change, and summer change respectively.

Shoal Zone Results: Annual

Over the last decade, the Shoal Zone net eroded for 6 years, and net accreted 4 years. The largest total accretion occurred between the Fall of 2016 survey and the Fall of 2017 survey, when the area gained ~5.93 million cubic feet of sediment. The largest erosive year occurred between the fall of 2017 and the fall of 2018 when the beach strand lost a total of 2,200,000 (+/- 1,000,000) cubic feet, losing an average of 2.85 feet of elevation per square foot. *(Figure 16)*

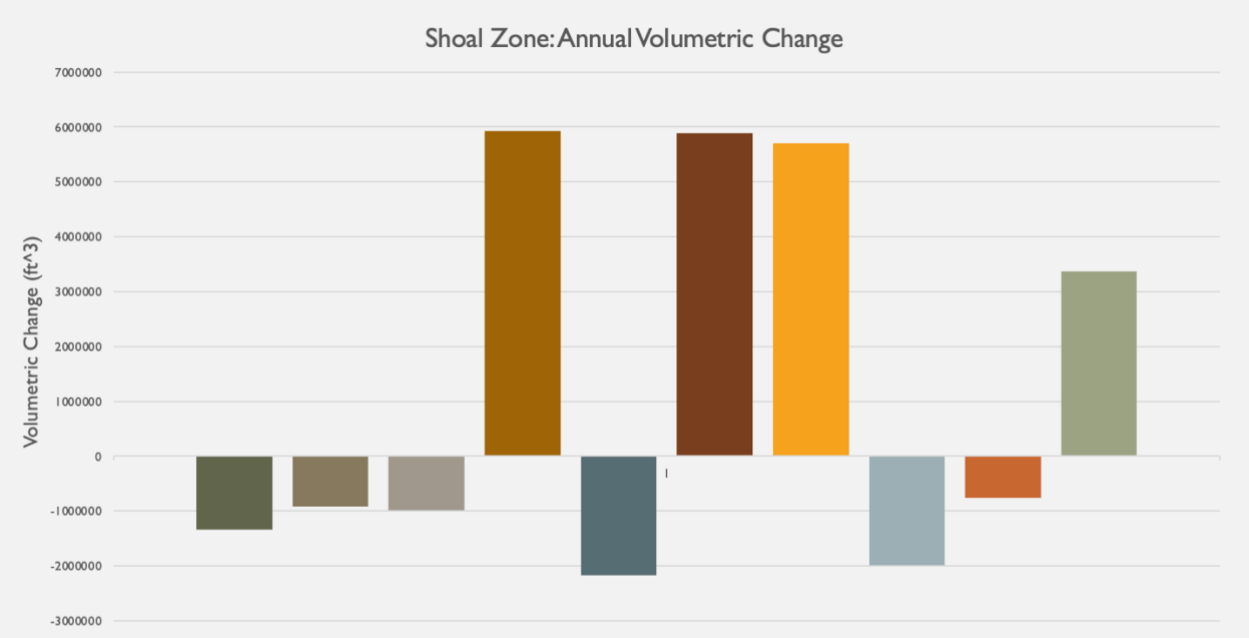

Figure 16: Beach Strand Zone annual volumetric change from 2014 - 2023

Shoal Zone Results: Seasonal

During the winter months, the Shoal Zone cumulatively accreted ~3.67 million cubic feet of sediment. Like all other zones, his relatively major long-term rate of accretion is determined by a period that exhibits extreme variance over time. Area 1 net accreted 4 winters, and net eroded 6 winters. Almost all the positive change occurred between the fall of 2017 and the fall of 2020, with only one year (2018) net eroding in that time frame. The largest change occurred in the winter of 2020 when the area gained a total of 4,300,000 (+/- 1,000,000) cubic feet During the summer months, Area 1 cumulatively gained ~6,300,000 (+/-1,000,000) cubic feet of sediment. The year of the greatest net change was from the spring of 2019 to the fall of 2019 when Area 1 accreted ~4,000,000 (+/- 1,000,000) cubic feet. *(Figure 17)*

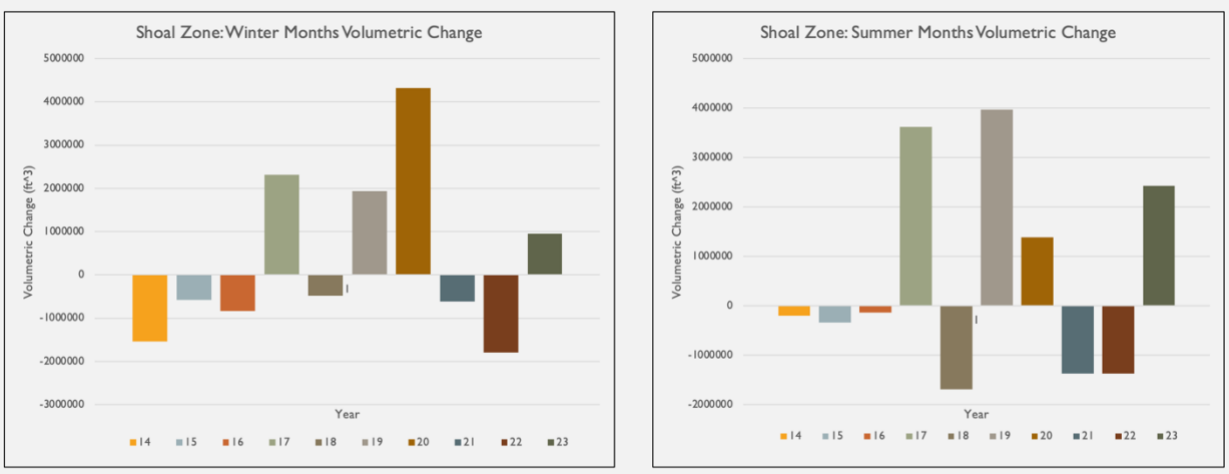

Figure 17: Shoal Zone winter season and summer season volumetric change respectively from 2014 to 2023

Data Error

There are a few major sources of error within the data. First, the time periods of surveys are not consistent, meaning the summer and winter analysis periods may not sum exactly to the year totals. Additionally, there is error within the single beam sonar survey method used which can be relatively high (DeWitt, 2013). On top of this, the changing inundation levels of smaller zones that consist of the shoreface may implicitly cause error due to the change in instrumentation from sonar to land based topography methods. Although the volumetric data displayed in this study seem large, during some seasons and years the total change of the beach strand was only a portion of a foot, meaning that even small instrument errors could lead to large volumetric error proportional to the error in the data analyzed in this paper.

Discussion

Zone 1 seasonal trends

While the Beach Strand data provides insight to the general trends of the area over different time periods, it does not demonstrate the variance that occurs within the zone. The results of this study demonstrated that erosion and accretion do not occur congruently across the Point of Emerald Isle. Rather there are seasonal hotspots of accretion and erosion within the area that influence the general trends of a given period. When compared to the Beach Strand Zone, Zone 1 showed extreme variance from the large-scale trends of the point. For example, Zone 1 net eroded significantly in the summertime over the last decade, representing an inverse pattern to the net accretional trend seen on the entirety of the beach strand. The point may have accreted

sand over the last decade; however, the distribution of that accretion is not even, and some areas may have net eroded during that time.

Hotspots for erosion and accretion move spatially throughout time, they do not occur in the same area every year. Stationary zones in this study provide insight to the idea that yearly and seasonal zoning of hotspots may be necessary to fully understand the distribution of the effects of environmental pressures. Hotspot migration was visualized using the raster difference maps which symbolizes the quantity of gain or loss over a period. Using these tools as visual aids, temporally driven spatial shifts in hotspots seemed to be random. However, more in depth analysis or modeling of how specific weather events interact with sediment transport patterns and the bathymetry of the area may provide more insight to hotspot movement.

Beach Strand Seasonal trends: Winter

Winter weather trends seem to support a net erosional effect over extended periods of time. Both waves and wind act as vector quantities and are the major mechanisms for nearshore current direction and intensity. The general weather pattern in emerald isle during the winter months is moderate N/NE wind, which promotes the longshore current to carry water and sediment east to west. Longshore current is a driver of sediment movement along the shoreline (Komar, 1983). This coupled with the offshore component of the wind aids in the net transport of sediment both alongshore to the west and offshore to the south. This pattern is magnified by extreme winter weather events, commonly referred to as nor'easters. Nor'easters create a strong flow of SE to SW wind in the area, which is then flipped to a dominant flow of NW to NE wind for periods of time that span for weeks or longer. These events have been observed to be

extremely erosive to bogue banks, which may be a result of the initial mobilization of sediment followed by the dominant transport away from the island. Additionally, nor'easters can be just as intense as tropical storms or hurricanes and are a much more frequent occurrence than tropical weather landfall. While there are periods of winter weather that may promote accretion on the point, the general patterns and major events that describe the winter season should lead to an erosive trend. Despite this, there was minor net accretion during the last decade. Variance in factors that influence the sediment dynamics of a given period is extremely high. No winter season is the same, and while there are established general patterns, the effects of major storms and the duration of erosive periods in the winter change every season. This coupled with anthropogenic change can lead to net accretion during the winter months.

Beach renourishment along Bogue Banks has been occurring for the last 40+ years and will continue to occur with the increasing rate of climate change impacts. Winter months are the ideal season for renourishments simply because there are significantly less people on the beach during that time frame. Despite the towns year-round population of roughly 2,300, during the summer there can be upwards of 60,000 people on the 13-mile beach strand of Emerald Isle alone (*Emerald Isle, NC | Official Website*, n.d.). During the 10-year period analyzed, there were two major renourishment events that took place on the point, both were post-storm renourishment efforts. The first took place in 2014 as a part of the "Post-Irene Renourishment" which added ~150,000 cubic feet to the Beach Strand Zone (*Monitoring | Carteret County, NC - Official Website*, 2020). The next took place in 2021 as a part of the "Post-Florence Renourishment" which added ~255,000 cubic feet to the Beach Strand Zone (*Monitoring | Carteret County, NC - Official Website*, 2020). Combined, the two renourishment operations account for less than 5% of the total accretion calculated between 2014 and 2023.

Renourishment also supplements the Beach Strand Zones accretional total in the winter, accounting for ~80% of the total change in the winter months over the last decade.

Renourishment increases the width of the beach and leads to more interaction between sandbars and the beach. This increase in interaction could lead to the area capitalizing on periods of accretion favoring weather. In turn, making winters when renourishment took place more likely to accrete further. This aspect of renourishment may act as a mechanism for post renourishment accretion spikes. While this paper does not investigate modeling of sediment dynamics, the data does show that renourishment years see net accretion which exceeds the total amount of sediment anthropogenically placed on the beach. In 2014, renourishment only accounted for ~9% of the total accretion and a similar pattern was seen in 2021 when renourishment accounted for ~6% of the total accretion. With this trend, future research should investigate the interaction between general winter conditions and the post renourishment accretion on the point of emerald isle.

Beach Strand Seasonal Trends: Summer

Summer weather trends seem to lead a net accretional effect over extended periods of time. The general weather pattern in emerald isle during the summer months is moderate SW to SE wind. This coupled with the onshore component of the wind aids in the net transport of sediment both alongshore to the east and onshore towards the north. The summer months, as defined in this study, accounted for ~95% of the total change over the last 10 years. However, accretion was not distributed evenly throughout time or space. Rather there was a significant amount of variability both between summer seasons and areas along the beach strand. Variability

can come from a multitude of factors in the summer months. First off, weather is unpredictable and, while there is an established predominant wind flow during those months, there is no way to exactly determine what the weather of subsequent summer seasons will be composed of.

Similar to the way that beach renourishment statistically offsets the general erosive trends of winter months, major tropical cyclone impacts during the summer months have the capacity to offset net accretion seen during other times in a given period and even reverse the periods net change to erosion. Throughout the ten analyzed summer seasons, three net eroded and one attributed roughly net zero change. Of these four total years of non-accretion, three of them were years of tropical cyclone landfall including Hurricane Arthur (2014), Hurricane Matthew (2016), and Hurricane Florence (2018). Additionally, the year of lowest net accretion was the year of near direct landfall by Tropical Storm Isais (2021).

Annual and Decadal Analysis of the Beach Strand

The beach strand accumulated 9,000,000 (+/- 600,000) cubic feet of sediment over the last ten years. Like the seasonal analysis, the distribution of the accretional trends of the beach strand were not congruent and the area displayed similar amounts of variance seen across other analysis periods. All of the results in this study demonstrate significant variance in the response to factors that influence the beach strand and suggest that the area is not a closed system. One piece of data that highlights this idea lies in the relationship between renourishment and volume in the winter months as it related to the decadal totals of accretion. All the beach renourishment that occurred within the period of analysis accounts for less than 5% of the total gain on the beach strand. This implicitly means that sand must be entering the Point system from elsewhere.

The data and discussion synthesized suggest a couple of hypotheses that may explain the impulse of sand into the area from exterior sources.

Proposed Mechanism: Shoals

After the analysis of the data of all zones on the beach, another analysis was set forth on the shoals on the eastern side of Bogue Inlet to investigate the possible relationship between shoal sediment flux and beach strand erosion or accretion. The idea behind the hypothesis was that the shoals of Bogue Inlet were acting as a trap that caught the sediment drifting westward from up beach areas due to the prevailing longshore current during winter months. Ultimately, leading to deposition onto the beach strand during subsequent summer southerly wind flow. Within this hypothesis, it was also suggested that renourishment of up beach areas may lead to more shoal accretion within the same year due to the influx of sand that is mobilized during renourishment.

Upon testing, it was discovered that the nearshore area accreted more than the beach strand, gaining nearly 11,000,000 (+/-1,000,000) million cubic feet of sediment over the last decade. The winter months accounted for 3.75 (+/- 1,000,000) cubic feet of gain and the summer months accounted for 6.5 (+/- 1,000,000) cubic feet of gain. Within the results, it was suggested that up beach renourishment did have a positive relationship with accretion along the shoals.

Additionally, there was no variance observed in the overall trend of accretion or erosion intra-annually, meaning that if it was a year of net erosion, both the summer and winter months net eroded within the Shoal Zone. This pattern is inverse to what is seen in the results from all areas on the beach strand which exhibited extreme variance intra-annually. contradictorily, the

variance inter-annually is higher than what is seen in the beach strand zone despite the shoal zone having less surface area than the beach strand.

Upon comparing the results of the beach strand to the results of the shoal zone, there is no significant alignment between the shoal erosion and beach accretion. The results from this study are ultimately inconclusive to whether the shoals act as a natural renourishment method for the point of Emerald Isle. Additionally, it was found that the shoal zone gained most of its sediment during the summer months like the beach strand. However, the zone also accreted significantly more during the winter when compared to the beach strand. All in all, a more focused project on the mechanisms that lead to the increased accumulation of sediment on the point in Emerald Isle when compared to up beach areas should be conducted. Some possible sources that should be studied include: a more in depth analysis of the shoals neighboring bogue inlet, estuarine-ocean sedimentation stemming from Bogue Sound, grain size studies conducted within Bogue Sound and its tributaries to study whether the grain size is conducive to bar concertation on the west side of the inlet, and surf zone modeling of sediment dynamics using the complex bathymetry and natural pressures of the area.

Conclusion

From analyzing multiple scales of temporal sedimentation patterns, to testing hypotheses that may cause these patterns, this study investigated many questions that were designed around the unusual accretionary trend on the point of Emerald Isle. Results showed that the beach strand does not contain the same sediment flux trends over time or space. Smaller zones on the beach

demonstrated that the beach strand does not change uniformly. Rather, different areas within the total zone may respond differently to constant environmental pressures. In terms of time scales, there was significant variance over the decade analysis. Not all years accreted despite the decadal trend of accretion, and not all seasons accreted even when the years encompassing them net accreted.

Over the decade of the analysis period, ~95% of the total accretion occurred within the summer months. This trend could be attributed to general summer weather patterns that drive sediment easterly, along and towards bogue banks. Oppositely, the winter months accounted for only ~5% of the net accretion. This trend can be accounted for by the general winter weather patterns and extreme winter weather events that drive sediment westerly, along and away from Bogue Banks. Counteracting winter weather trends are anthropogenic inputs to the beach strand via renourishment. These inputs account for less than 5% of the total accretion seen between 2014 and 2023 and suggest the idea that the point of Emerald Isle is far from a closed system. Sediment has to be coming from other sources to account for the over ~9 million cubic feet of sand that has been naturally accreted there over the last 10 years, and this study provides insight on a couple hypotheses that should be focused on in future research.

One of these hypotheses was tested, it focused on the shoals east of bogue inlet acting as a system that could retain sediment flowing westward during the winter months and then naturally renourish the beach strand zone during subsequent times off onshore winds in the summer. The analysis conducted proved to be inconclusive. However, some interesting findings include the major decadal trend of accretion, the lack of seasonal variance when compared to the beach strand area, and the increase in year-to-year variance. All these ideas could lead future

research on the mechanism of the ebb tide delta and shoal zone of Bogue Inlet, especially investigating its possible interaction with the natural renourishment of the beach strand.

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Appendix

Figure 1 – Hurricane Florence track, damage, and aerial image upon landfall

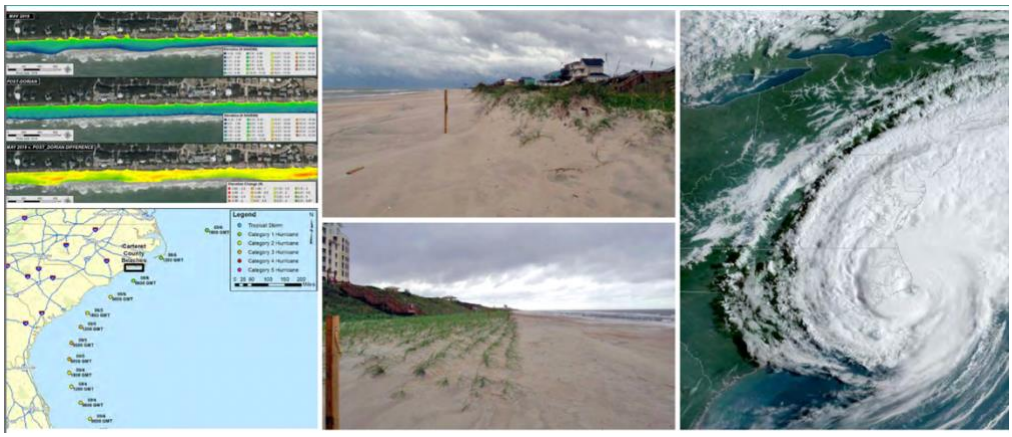

Figure 2 – Hurricane Dorian track, damage, and aerial image upon landfall

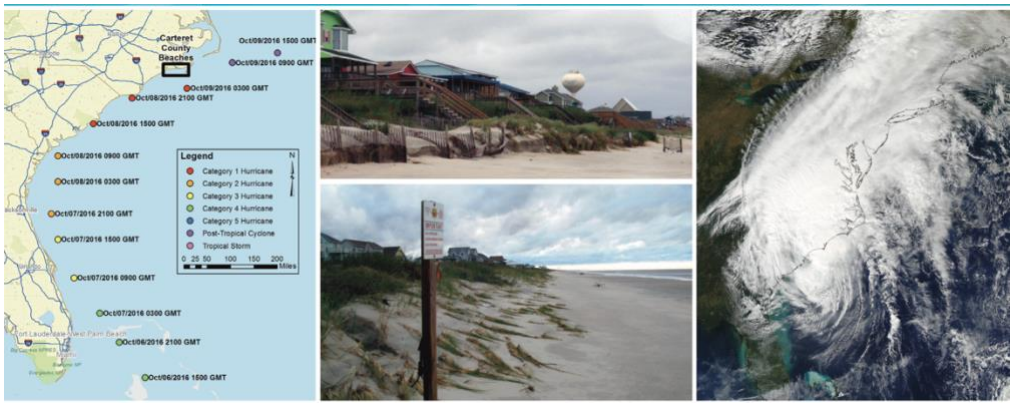

Figure 3 – Hurricane Matthew track, damage, and aerial image upon landfall